Wednesday, 25th March, 2026

Assignment on postulates of quantum mechanics

Wavefunction postulate

① State the wavefunction postulate of quantum mechanics.

$\boxed{\text{Wavefunction lives in Hilbert space}}$

- The state of quantum system is completely described by a wavefunction $\psi(x,t)$.
- In the Hilbert space, the wavefunction is represented by a ket $|\psi\rangle$.
- The bra $\langle\psi|$ is the dual of the ket $|\psi\rangle$ and bra $\langle\psi|$ resides in the dual space corresponding to Hilbert space $\mathcal{H}$.
- The ket $|\psi\rangle$ is normalized i.e. the norm $\langle\psi|\psi\rangle = 1$
- The inner product of ket $|\psi\rangle$ with ket $|\phi\rangle$ is given by

$$\boxed{\langle\phi|\psi\rangle = \langle\psi|\phi\rangle^*}$$

- Matrix representation of $|\psi\rangle$ is a d-dimensional column vector in a d- dimensional Hilbert space.

- If $\{|i\rangle\}$ is the computational basis then

$$|\psi\rangle = \sum_i c_i |i\rangle \text{, where } c_i = \langle i|\psi\rangle$$

- $\psi(x,t)$ is the projection of $|\psi(t)\rangle$ onto the coordinate basis $\{|x\rangle\}$

i.e. $\langle x|\psi(t)\rangle =: \psi(x,t)$

- $\psi(x,t)$ is
  - single valued
  - finite
  - continuous
  - normalizable
- The probability density of finding particle is $|\psi(x,t)|^2$ so that probability of finding particle between x and $x+dx$ is $|\psi(x,t)|^2 dx$

② Concept of *normalized* kets

(a) Consider a collimated beam of light with a given state of polarization as shown in figure.1. If the light is propagating along $+y$ with amplitude E_0, wavenumber K, and angular frequency ω, then

$$|H\rangle = E_0 \, \hat{i} \exp(Ky - \omega t)$$
$$|V\rangle = E_0 \hat{k} \exp(Ky - \omega t)$$

are the independent horizontal and vertical states of polarization respectively.

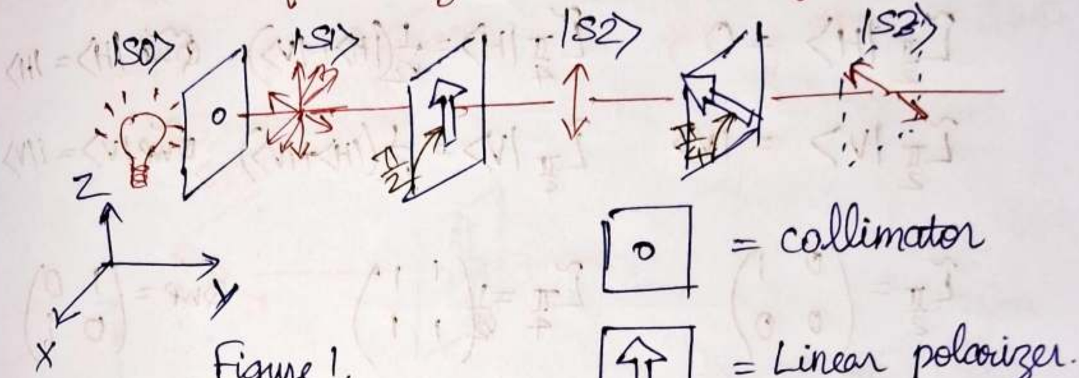


Figure 1.

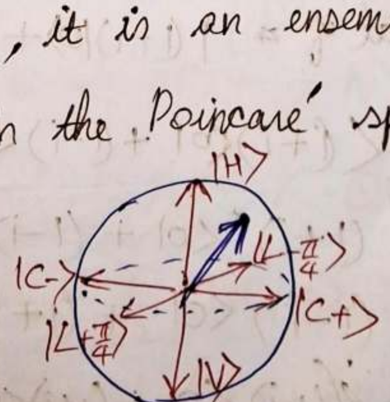
Find the normalized ket $|S3\rangle$ representing the state of polarization after passing through

- collimator $(|S0\rangle \rightarrow |S1\rangle)$
- linear polarizer $\{\tilde{L}_{\frac{\pi}{2}}\}$ $(|S1\rangle \rightarrow |S2\rangle)$
- linear polarizer $\tilde{L}_{\frac{\pi}{4}}$ $(|S2\rangle \rightarrow |S3\rangle)$

Here $\tilde{L}_{\frac{\pi}{2}}$ is along $\hat{z}$, $\tilde{L}_{\frac{\pi}{4}}$ makes $\frac{\pi}{4}$ $\bar{c}$ x-axis

- collimator

After collimation $|S1\rangle$ is unpolarized. Therefore, it is an ensemble of all the states on the Poincaré sphere.



Note: $|H\rangle$ and $|V\rangle$ are located on the north and south pole respectively.

So $|S1\rangle = \cos\frac{\theta}{2}\,|H\rangle + e^{i\phi}\sin\frac{\theta}{2}\,|V\rangle$

where

$|H\rangle, |V\rangle$ are linearly independent horizontal and vertical states of polarization respectively,

θ and ϕ are latitude and longitude of the polarization on the Poincaré sphere[1].

Unpolarized light consists of all the polarization states i.e.

$0 \leq \theta \leq \pi$, and $0 \leq \phi \leq 2\pi$.

- linear polarizer $\tilde{L}_{\frac{\pi}{2}}$

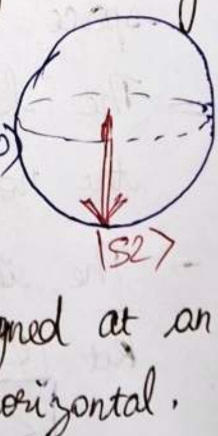
The polarizer filters out the polarization state.

Since the polarizer is linear,

$|S2\rangle = \cos\alpha\,|H\rangle + \sin\alpha\,|V\rangle$

Now since polarizer is aligned along the vertical direction, $\alpha = \frac{\pi}{2}$

$\Rightarrow |S2\rangle = |V\rangle \Rightarrow (\theta, \phi) = (\pi, 0)$



- linear polarizer $\tilde{L}_{\frac{\pi}{4}}$

The second polarizer is aligned at an angle $\pi/4$ away from the horizontal.

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[1] Henri Poincaré was a French mathematician, theoretical physicist. He was described as the "Last Universalist". The acute e (é) is spelled as ae.

Hence, $\alpha = \frac{\pi}{4}$

$\therefore \; |S3\rangle = \frac{1}{\sqrt{2}}|H\rangle + \frac{1}{\sqrt{2}}|V\rangle$

$\Rightarrow (\theta, \phi) = \left(\frac{\pi}{2}, 0\right)$

Furthermore, the beam is now passed through a quarter-wave plate (QWP) as shown in Figure 2. Find the normalized ket $|S4\rangle$ representing the state of polarization after passing through QWP ($|S3\rangle \to |S4\rangle$)

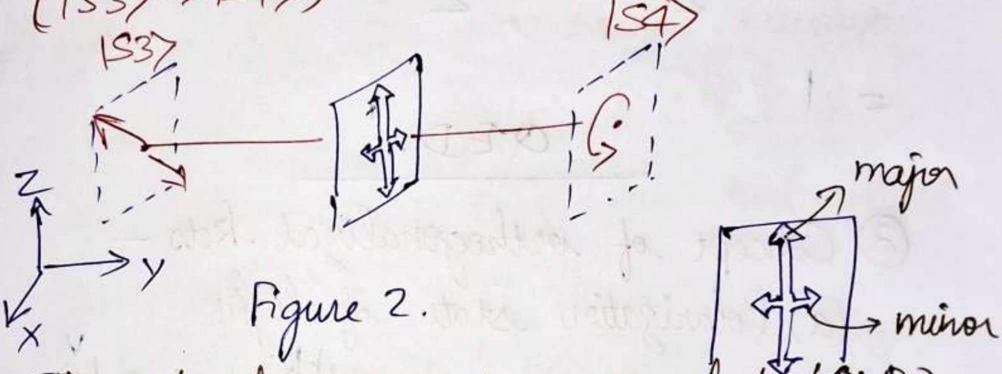

Figure 2.

The role of a quarter wave plate (QWP) is to advance the phase of polarization component along the major axis of QWP by $\frac{\pi}{2}$.

So Here, the major axis is along $|V\rangle$

$\Rightarrow$ after the $\left[\widetilde{QWP}, \; |V\rangle \to e^{i\frac{\pi}{2}}|V\rangle\right]$

$\left[\widetilde{QWP}|H\rangle = |H\rangle\right]$

$|S3\rangle = \frac{1}{\sqrt{2}}|H\rangle + \frac{1}{\sqrt{2}}|V\rangle$

$\Rightarrow |S4\rangle = \cancel{\frac{1}{\sqrt{2}}|H\rangle +} \; \widetilde{QWP}\,|S3\rangle$

$= \frac{1}{\sqrt{2}}\widetilde{QWP}|H\rangle + \frac{1}{\sqrt{2}}\widetilde{QWP}|V\rangle$

$= \frac{1}{\sqrt{2}}\left(|H\rangle + i|V\rangle\right)$

Now, that we know the action of each of polarizers on the linearly independent polarizations states $|H\rangle$ and $|V\rangle$, we can construct their matrix representations

$\tilde{L}_{\frac{\pi}{2}}$ $\tilde{L}_{\frac{\pi}{4}}$ $\tilde{QWP}$

$\tilde{L}_{\frac{\pi}{2}}|H\rangle = 0$ $\tilde{L}_{\frac{\pi}{4}}|H\rangle = \frac{1}{\sqrt{2}}(|H\rangle + |V\rangle)$ $\tilde{QWP}|H\rangle = |H\rangle$

$\tilde{L}_{\frac{\pi}{2}}|V\rangle = |V\rangle$ $\tilde{L}_{\frac{\pi}{4}}|V\rangle = \frac{1}{\sqrt{2}}(|H\rangle + |V\rangle)$ $\tilde{QWP}|V\rangle = i|V\rangle$

$\tilde{L}_{\frac{\pi}{2}} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ $\tilde{L}_{\frac{\pi}{4}} = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ $\tilde{QWP} = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}$

(b) spin-$\frac{1}{2}$ system

(i) Normalize the ket

$$|\psi\rangle = (1+i)|0\rangle + (1-i)|1\rangle$$

$\langle\psi|\psi\rangle = ?$

$\langle\psi|$ ~~is~~ is dual of $|\psi\rangle$

$= (1+i)|0\rangle + (1-i)|1\rangle$

$= |(1+i)|0\rangle + |(1-i)|1\rangle\rangle$

$\langle\psi|$ is dual of ~~=~~ $|(1+i)|0\rangle + (1-i)|1\rangle\rangle$

$\Rightarrow \langle\psi| = \langle(1+i)\langle 0| + (1-i)\langle 1||$

$= (1+i)^*\langle 0| + (1-i)^*\langle 1|$

$= (1-i)\langle 0| + (1+i)\langle 1|$

So $\langle\psi|\psi\rangle = ((1-i)\langle 0| + (1+i)\langle 1|)((1+i)|0\rangle + (1-i)|1\rangle)$

$$= (1-i^2)\cancel{2}\langle 0|0\rangle + (1-i)^2\cancel{\langle 0|1\rangle}$$
$$+ (1+i)^2\cancel{\langle 1|0\rangle} + (1-i^2)\langle 1|1\rangle$$
$$= 2 + 2$$
$$\langle\psi|\psi\rangle = 4$$

$\therefore$ normalized $|\psi\rangle$ is

$$\frac{|\psi\rangle}{\sqrt{\langle\psi|\psi\rangle}} = \frac{1}{2}\left[(1+i)|0\rangle + (1-i)|1\rangle\right]$$

(ii) Prove that the parametrized waveket
$$|\psi(\theta,\phi)\rangle = \cos\frac{\theta}{2}|0\rangle + e^{i\phi}\sin\frac{\theta}{2}|1\rangle$$
has norm 1. Here θ, ϕ are real valued parameters.

$$|\psi(\theta,\phi)\rangle = \cos\frac{\theta}{2}|0\rangle + e^{i\phi}\sin\frac{\theta}{2}|1\rangle$$
$$\langle\psi|\psi\rangle = ?$$

$\langle\psi(\theta,\phi)|$ is dual of $|\psi(\theta,\phi)\rangle$
$$= \cos\frac{\theta}{2}|0\rangle + e^{i\phi}\sin\frac{\theta}{2}|1\rangle$$
$$= |\cos\frac{\theta}{2}|0\rangle + e^{i\phi}\sin\frac{\theta}{2}|1\rangle\rangle$$
$$\Rightarrow \langle\psi(\theta,\phi)| = \langle\cos\frac{\theta}{2}\langle 0| + e^{i\phi}\sin\frac{\theta}{2}\langle 1||$$
$$= \left(\cos\frac{\theta}{2}\right)^*\langle 0| + \left(e^{i\phi}\sin\frac{\theta}{2}\right)^*\langle 1|$$
$$= \cos\frac{\theta}{2}\langle 0| + e^{-i\phi}\sin\frac{\theta}{2}\langle 1|$$

// $(e^{i\phi})^* = (\cos\phi + i\sin\phi)^* = \cos\phi - i\sin\phi$
$$= \cos(-\phi) + i\sin(-\phi)$$
$$= e^{-i\phi}$$

$\therefore \langle \psi(\theta,\phi) | \psi(\theta,\phi) \rangle$

$$= \left(\cos\frac{\theta}{2}\langle 0| + e^{-i\phi}\sin\frac{\theta}{2}\langle 1|\right)\left(\cos\frac{\theta}{2}|0\rangle + e^{+i\phi}\sin\frac{\theta}{2}|1\rangle\right)$$

$$= \begin{cases} \cos^2\frac{\theta}{2}\langle 0|0\rangle + e^{+i\phi}\cos\frac{\theta}{2}\sin\frac{\theta}{2}\cancel{\langle 0|1\rangle} \\ + e^{-i\phi}\cos\frac{\theta}{2}\sin\frac{\theta}{2}\cancel{\langle 1|0\rangle} + e^{-i\phi}e^{+i\phi}\sin^2\frac{\theta}{2}\langle 1|1\rangle \end{cases}$$

$$= \cos^2\frac{\theta}{2} + \sin^2\frac{\theta}{2}$$

$$= 1$$

Q.E.D.[1]

③ Concept of orthogonalized kets –

(a) Polarization states of light

(i) If $|L_\theta\rangle = \cos\theta\,|\overset{H}{\leftrightarrow}\rangle + \sin\theta\,|\overset{V}{\updownarrow}\rangle$, then P.T. $|L_{+\frac{\pi}{4}}\rangle$ is orthogonal to $|L_{-\frac{\pi}{4}}\rangle$

$$|L_{+\frac{\pi}{4}}\rangle = \cos\frac{\pi}{4}|H\rangle + \sin\frac{\pi}{4}|V\rangle$$

$$|L_{+\frac{\pi}{4}}\rangle = \frac{1}{\sqrt{2}}|H\rangle + \frac{1}{\sqrt{2}}|V\rangle$$

$$|L_{-\frac{\pi}{4}}\rangle = \cos\left(-\frac{\pi}{4}\right)|H\rangle + \sin\left(-\frac{\pi}{4}\right)|V\rangle$$

$$|L_{-\frac{\pi}{4}}\rangle = \frac{1}{\sqrt{2}}|H\rangle - \frac{1}{\sqrt{2}}|V\rangle$$

so that

$$\langle L_{+\frac{\pi}{4}} | L_{-\frac{\pi}{4}}\rangle = \left(\frac{1}{\sqrt{2}}\langle H| + \frac{1}{\sqrt{2}}\langle V|\right)\left(\frac{1}{\sqrt{2}}|H\rangle - \frac{1}{\sqrt{2}}|V\rangle\right)$$

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1. Q.E.D. = quod erat demonstrandum (Latin for "that which was to be demonstrated)

$$= \frac{1}{2}\langle H|H\rangle - \frac{1}{2}\cancel{\langle H|V\rangle} + \frac{1}{2}\cancel{\langle V|H\rangle} - \frac{1}{2}\langle V|V\rangle$$

$$= \frac{1}{2} - \frac{1}{2}$$

$$\langle L_{\frac{\pi}{4}} | L_{-\frac{\pi}{4}}\rangle = 0$$

$\Rightarrow$ The kets $|L_{+\frac{\pi}{4}}\rangle$ and $|L_{-\frac{\pi}{4}}\rangle$ are orthogonal to each other.

(ii) If $|C_+\rangle = \frac{1}{\sqrt{2}}(|H\rangle + i|V\rangle)$, then find the circularly polarized ket $|C_-\rangle$ that is orthogonal to $|C_+\rangle$.

Let $|C_-\rangle = \alpha|H\rangle + \beta|V\rangle$ with $|\alpha|^2 + |\beta|^2 = 1$

Since $|C_-\rangle$ is orthogonal to $|C_+\rangle$

$$\langle C_+|C_-\rangle = 0 \Leftrightarrow \left\langle \frac{1}{\sqrt{2}}(\langle H| + i\langle V|) \,\middle|\, \alpha|H\rangle + \beta|V\rangle \right\rangle = 0$$

$$\Leftrightarrow \frac{1}{\sqrt{2}}(\langle H| - i\langle V|)\cdot(\alpha|H\rangle + \beta|V\rangle) = 0$$

$$\Leftrightarrow \alpha - i\beta = 0$$

$$\Leftrightarrow \alpha = i\beta$$

$$\Leftrightarrow \beta = -i\alpha$$

WLG (Without loss of generalization)

let $\alpha = 1$

$\Rightarrow \beta = -i$,

But $|\alpha|^2 + |\beta|^2 = 1 \Rightarrow \alpha = \frac{1}{\sqrt{2}}, \beta = \frac{-i}{\sqrt{2}}$

$\therefore |C_-\rangle = \frac{1}{\sqrt{2}}(|H\rangle - i|V\rangle)$

(b) spin-$\frac{1}{2}$ system

(i) A general ket for a spin-$\frac{1}{2}$ system is given by the two parameter form

$$|\psi(\theta,\phi)\rangle = \cos\frac{\theta}{2}|0\rangle + e^{i\phi}\sin\frac{\theta}{2}|1\rangle$$

Prove that $|\psi(\pi-\theta, \pi+\phi)\rangle$ is orthogonal to $|\psi(\theta,\phi)\rangle$.

$$|\psi(\theta,\phi)\rangle = \cos\frac{\theta}{2}|0\rangle + e^{i\phi}\sin\frac{\theta}{2}|1\rangle$$

$$|\psi(\pi-\theta,\pi+\phi)\rangle = \cos\left(\frac{\pi-\theta}{2}\right)|0\rangle + e^{i(\pi+\phi)}\sin\left(\frac{\pi-\theta}{2}\right)|1\rangle$$

$$= \sin\frac{\theta}{2}|0\rangle + e^{i\pi}e^{i\phi}\cos\frac{\theta}{2}|1\rangle$$

$$= \sin\frac{\theta}{2}|0\rangle - e^{i\phi}\cos\frac{\theta}{2}|1\rangle$$

so that

$$\langle\psi(\pi-\theta,\pi+\phi)|\psi(\theta,\phi)\rangle$$

$$= \left(\sin\frac{\theta}{2}\langle 0| + e^{i\phi}\cos\frac{\theta}{2}\langle 1|\right)\left(\cos\frac{\theta}{2}|0\rangle + e^{i\phi}\sin\frac{\theta}{2}|1\rangle\right)$$

$$= \sin\frac{\theta}{2}\cos\frac{\theta}{2}\langle 0|0\rangle + e^{i\phi}\sin\frac{\theta}{2}\langle 0|1\rangle$$
$$+ e^{i\phi}\cos^2\frac{\theta}{2}\langle 1|0\rangle + e^{i2\phi}\sin\frac{\theta}{2}\cos\frac{\theta}{2}\langle 1|1\rangle$$

Observable postulate

① State the observable postulate of quantum mechanics

$\boxed{\text{An observable is represented by Hermitian operator}}$

- An observable $q(x, p)$ is associated with an operator $\tilde{Q}$ such that the measurement outcomes of q are eigenvalues of $\tilde{Q}$.
- The expectation value of q in the discrete computational basis $\{|i\rangle\}$ is given by

$$\boxed{\langle \tilde{Q} \rangle = \langle \psi | \tilde{Q} | \psi \rangle,}$$

where $|\psi\rangle$ is the state of the quantum system

- In the continuous coordinate basis $\{|x\rangle\}$, the expectation value of q is given by

$$\boxed{\langle q \rangle = \int \psi^*(x)\, Q\left(x, \frac{\hbar}{i}\frac{\partial}{\partial x}\right) \psi(x)\, dx}$$

where $\psi(x)$ is the projection of $|\psi\rangle$ onto $|x\rangle$ i.e.

$$\langle x | \psi \rangle =: \psi(x)$$

② From action of observable to operator -

(a) spin - $\frac{1}{2}$ system

(i) The observable ~~has~~ $\tilde{S}_z$ has the following action on the computational basis $\{|0\rangle, |1\rangle\}$

$$\tilde{S}_z|0\rangle = \frac{\hbar}{2}|0\rangle$$

$$\tilde{S}_z|1\rangle = -\frac{\hbar}{2}|1\rangle$$

Find the matrix representation of operator $\tilde{S}_z$.

$$\tilde{S}_z\begin{pmatrix}1\\0\end{pmatrix} = \frac{\hbar}{2}\begin{pmatrix}1\\0\end{pmatrix}$$

$$\tilde{S}_z\begin{pmatrix}0\\1\end{pmatrix} = \frac{\hbar}{2}\begin{pmatrix}0\\-1\end{pmatrix}$$

From the action of $\tilde{S}_z$ on the computational basis, the matrix representation is given by

$$\tilde{S}_z = \frac{\hbar}{2}\begin{pmatrix}1 & 0\\0 & -1\end{pmatrix}$$

(ii) The observable $\tilde{S}_x$ has the following actions

$$\tilde{S}_x|0\rangle = \frac{\hbar}{2}|1\rangle$$

$$\tilde{S}_x|1\rangle = \frac{\hbar}{2}|0\rangle$$

Find the matrix representation of operator $\tilde{S}_x$

$$\tilde{S}_x\begin{pmatrix}1\\0\end{pmatrix} = \frac{\hbar}{2}\begin{pmatrix}0\\1\end{pmatrix}$$

$$\tilde{S}_x\begin{pmatrix}0\\1\end{pmatrix} = \frac{\hbar}{2}\begin{pmatrix}1\\0\end{pmatrix}$$

$$\therefore\ \tilde{S}_x = \frac{\hbar}{2}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

(3) From operator to action of observable

(a) spin-$\frac{1}{2}$ system

(i) The matrix representation of operator corresponding to the observable $\hat{S}_y$ is

$$\tilde{S}_y = \frac{\hbar}{2}\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}.$$

Find the action of $\tilde{S}_y$ on the computational basis.

Action of $\tilde{S}_y$ on computational basis

$$\left\{ \underset{|0\rangle}{\begin{pmatrix} 1 \\ 0 \end{pmatrix}}, \underset{|1\rangle}{\begin{pmatrix} 0 \\ 1 \end{pmatrix}} \right\} \text{ is}$$

$$\tilde{S}_2 |0\rangle = \frac{\hbar}{2}\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$= \frac{\hbar}{2}\begin{pmatrix} 0 \\ i \end{pmatrix}$$

$$\tilde{S}_2 |0\rangle = i\frac{\hbar}{2} |1\rangle$$

and $\tilde{S}_2 |1\rangle = \frac{\hbar}{2}\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}\begin{pmatrix} 0 \\ 1 \end{pmatrix}$

$$= \frac{-i\hbar}{2}\begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$\tilde{S}_2 |1\rangle = -i\frac{\hbar}{2} |0\rangle$$

④ From operator to eigenvalues and eigenkets —

(a) spin-$\frac{1}{2}$ system

(i) Find the eigenvalues and eigenkets of the observables $\tilde{S}_z$ and $\tilde{S}_x$ whose matrix representations are given by

$$\tilde{S}_z = \frac{\hbar}{2}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \tilde{S}_x = \frac{\hbar}{2}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$\tilde{S}_z = \frac{\hbar}{2}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \qquad \tilde{S}_x = \frac{\hbar}{2}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

eigenvalues problem

$$\tilde{S}_z \vec{x} = \lambda \vec{x} \qquad \tilde{S}_x \vec{x} = \lambda \vec{x}$$

$$\Leftrightarrow \det(\tilde{S}_z - \lambda) = 0 \qquad \Leftrightarrow \det(\tilde{S}_x - \lambda) = 0$$

$$\Leftrightarrow \begin{vmatrix} \frac{\hbar}{2}-\lambda & 0 \\ 0 & -\frac{\hbar}{2}-\lambda \end{vmatrix} = 0 \qquad \Leftrightarrow \begin{vmatrix} -\lambda & \frac{\hbar}{2} \\ \frac{\hbar}{2} & -\lambda \end{vmatrix} = 0$$

$$\Leftrightarrow (\lambda - \tfrac{\hbar}{2})(\lambda + \tfrac{\hbar}{2}) = 0 \qquad \Leftrightarrow \lambda^2 - \frac{\hbar^2}{4} = 0$$

$$\Leftrightarrow \lambda = \pm\frac{\hbar}{2} \qquad \Leftrightarrow \lambda = \pm\frac{\hbar}{2}$$

eigenkets

$$\lambda = \frac{\hbar}{2} \qquad\qquad \lambda = \frac{\hbar}{2}$$

$$\Rightarrow \begin{pmatrix} 0 & 0 \\ 0 & -\hbar \end{pmatrix}\begin{pmatrix} \alpha \\ \beta \end{pmatrix} = 0, \qquad \Rightarrow \begin{pmatrix} -\frac{\hbar}{2} & \frac{\hbar}{2} \\ \frac{\hbar}{2} & -\frac{\hbar}{2} \end{pmatrix}\begin{pmatrix} \alpha \\ \beta \end{pmatrix} = 0$$

$$\Rightarrow \beta = 0$$

$$\Rightarrow \alpha = \beta$$

$$\Rightarrow \left|+\tfrac{\hbar}{2}\right\rangle_z = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$\Rightarrow \left|+\tfrac{\hbar}{2}\right\rangle_x = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$\lambda = -\frac{\hbar}{2}$

$$\Rightarrow \begin{pmatrix} \hbar & 0 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} \alpha \\ \beta \end{pmatrix} = 0$$

$$\Rightarrow \alpha = 0$$

$$\Rightarrow |-\tfrac{\hbar}{2}z\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$\lambda = -\frac{\hbar}{2}$

$$\Rightarrow \begin{pmatrix} \frac{\hbar}{2} & \frac{\hbar}{2} \\ \frac{\hbar}{2} & \frac{\hbar}{2} \end{pmatrix}\begin{pmatrix} \alpha \\ \beta \end{pmatrix} = 0$$

$$\Rightarrow \alpha + \beta = 0$$

$$\Rightarrow |-\tfrac{\hbar}{2}x\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

(4) From eigenkets and eigenvalues to operator

(a) spin-$\frac{1}{2}$ system

(i) The eigenvalues of observable $\tilde{S}_y$ are $\pm\frac{\hbar}{2}$, and eigenkets are

$$|\pm\tfrac{1}{2}y\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ \pm i \end{pmatrix}$$

Find the matrix representation of observable $\tilde{S}_y$.

Matrix representation $\bar{c}$ eigenkets and eigenvalues is given by

$$\tilde{Q} = \sum_i q_i |q_i\rangle\langle q_i|$$

where q_i is the eigenvalue corresponding to eigenket $|q_i\rangle$

So $$\tilde{S}_y = \frac{\hbar}{2}|+\tfrac{1}{2}y\rangle\langle +\tfrac{1}{2}y| - \frac{\hbar}{2}|\tfrac{-1}{2}y\rangle\langle \tfrac{-1}{2}y|$$

$$= \frac{\hbar}{2}\left[\frac{1}{\sqrt{2}}\frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ i \end{pmatrix}(1\ -i) - \frac{1}{\sqrt{2}}\frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -i \end{pmatrix}(1\ +i)\right]$$

$$= \frac{\hbar}{2}\frac{1}{2}\left[\begin{pmatrix} 1 & -i \\ +i & 1 \end{pmatrix} - \begin{pmatrix} 1 & i \\ -i & 1 \end{pmatrix}\right].$$

$$= \frac{\hbar}{4}\begin{pmatrix} 0 & -2i \\ +2i & 0 \end{pmatrix} = \frac{\hbar}{2}\begin{pmatrix} 0 & -i \\ +i & 0 \end{pmatrix}$$

Time evolution postulate

① State the time-evolution postulate of quantum mechanics.

- The ket $|\psi\rangle$ evolves in time as per the Schrödinger equation

$$i\hbar \frac{d}{dt}|\psi\rangle = \tilde{\mathcal{H}}|\psi\rangle,$$

where $\tilde{\mathcal{H}}$ is the Hamiltonian operator given by

$$\tilde{\mathcal{H}} = \tilde{T} + \tilde{V}$$

$$= \frac{\tilde{P}^2}{2m} + \tilde{V}$$

In the coordinate basis representation

$$\tilde{\mathcal{H}} \rightarrow H(x) = -\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + V(x)$$

- If the eigenvalues of $\tilde{\mathcal{H}}$ are E_i corresponding to energy eigenkets $|E_i\rangle$, then the ~~was~~ ket $|\psi(0)\rangle$ at time $t=0$ evolves in time as

$$|\psi(0)\rangle = \sum_i c_i(0)|E_i\rangle$$

$$|\psi(t)\rangle = \sum_i e^{-\frac{iE_i t}{\hbar}} c_i(0)|E_i\rangle$$

② The Hamiltonian for a certain three-level system is represented by the matrix

$$\tilde{H} = \begin{pmatrix} a & 0 & b \\ 0 & c & 0 \\ b & 0 & a \end{pmatrix},$$

where a, b, and c are real numbers.

(a) If the system starts out in the state

$$|S(0)\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix},$$

what is $|S(t)\rangle$?

Consider the Hamiltonian

$$\tilde{H} = \begin{pmatrix} a & 0 & b \\ 0 & c & 0 \\ b & 0 & a \end{pmatrix}$$

Here, the computational basis is $\{|0\rangle, |1\rangle, |2\rangle\}$

In the subspace spanned by $|1\rangle$, the Hamiltonian is diagonal and has no coupling with other axes; thus, the eigenvalue is $c = E_1$ with eigenket $|1\rangle$.

i.e. $\tilde{H}|1\rangle = c|1\rangle$ // Here $E_1 = c$

Now $|S(0)\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = |1\rangle$

So initial state is stationary state as it is eigenket of Hamiltonian.

Therefore $|S(t)\rangle = e^{-i\frac{E_1 t}{\hbar}}|1\rangle = e^{\frac{-ict}{\hbar}}|1\rangle$

(b) If the system starts out in the state

$$|S(0)\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix},$$

what is $|S(t)\rangle$?

Now, since the starting state belongs to the subspace spanned by $|0\rangle$ and $|2\rangle$ in which the ~~Hamiltonian~~ corresponding block of Hamiltonian is non-diagonal i.e

$$\text{block-}\tilde{H} \doteq \begin{pmatrix} a & b \\ b & a \end{pmatrix},$$

we first need to diagonalize the $\tilde{H}$ in $\{|0\rangle - |2\rangle\}$ sub-space

$$\lambda = a \pm b$$

and $|a \pm b\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ \pm 1 \end{pmatrix}$

So the matrix representation of these eigenkets in the complete basis is

$$|a \pm b\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 0 \\ \pm 1 \end{pmatrix}$$

Let us decompose the starting state in the energy eigenbasis

$$|S(0)\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}}\left(|a+b\rangle + |a-b\rangle\right)$$

so that

$$|S(t)\rangle = \frac{1}{\sqrt{2}}\left[e^{-\frac{i(a+b)t}{\hbar}}|a+b\rangle + e^{-\frac{i(a-b)t}{\hbar}}|a-b\rangle\right]$$

$$= \frac{1}{\sqrt{2}}\frac{1}{\sqrt{2}}\left[e^{-i\frac{(a+b)t}{\hbar}}\begin{pmatrix}1\\0\\1\end{pmatrix} + e^{-i\frac{(a-b)t}{\hbar}}\begin{pmatrix}1\\0\\-1\end{pmatrix}\right]$$

$$= \frac{1}{2}\begin{pmatrix} e^{-\frac{i(a+b)t}{\hbar}} + e^{-\frac{i(a-b)t}{\hbar}} \\ 0 \\ e^{-i\frac{(a+b)t}{\hbar}} - e^{-\frac{i(a-b)t}{\hbar}} \end{pmatrix}$$

$$= \frac{1}{2}\begin{pmatrix} e^{-\frac{iat}{\hbar}}\left(e^{-\frac{ibt}{\hbar}} + e^{+\frac{ibt}{\hbar}}\right) \\ 0 \\ e^{-\frac{iat}{\hbar}}\left(e^{-\frac{ibt}{\hbar}} - e^{-\frac{ibt}{\hbar}}\right) \end{pmatrix}$$

$$= \frac{1}{2}\cdot 2\begin{pmatrix} e^{-\frac{iat}{\hbar}}\cos\left(\frac{bt}{\hbar}\right) \\ 0 \\ -ie^{-\frac{iat}{\hbar}}\sin\left(\frac{bt}{\hbar}\right) \end{pmatrix}$$

$$\therefore\ |S(t)\rangle = \begin{pmatrix} e^{-\frac{iat}{\hbar}}\cos\left(\frac{bt}{\hbar}\right) \\ 0 \\ -ie^{-\frac{iat}{\hbar}}\sin\left(\frac{bt}{\hbar}\right) \end{pmatrix}$$

Measurement postulate

① State the measurement postulate

A measurement of observable on a quantum system $|\psi\rangle$ collapses it into an eigenstate $|\phi\rangle$ of the observable with probability $P_\phi = |\langle\phi|\psi\rangle|^2$.

- This is called the Copenhagen interpretation of quantum mechanics.
Also called as collapse of the wavefunction
It is also called the Born rule.

② The Hamiltonian for a certain three-level system is represented by the matrix

$$\tilde{H} = \hbar\omega \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

Two other observables, A and B, represented by the matrices

$$\tilde{A} = \lambda \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 2 \end{pmatrix}, \quad \tilde{B} = \mu \begin{pmatrix} 2 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix},$$

where ω, λ, and μ are positive real numbers.

(a) Find the eigenvalues ~~of~~ and eigenkets of $\tilde{H}$, $\tilde{A}$, and $\tilde{B}$.

$\tilde{H}$ is already diagonal in the computational basis $\{|0\rangle, |1\rangle, |2\rangle\}$

Hence eigenvalues of $\tilde{H}$ are $\hbar\omega$, $2\hbar\omega$, $2\hbar\omega$

and eigenkets are $|\hbar\omega\rangle = |0\rangle$, $|2\hbar\omega\rangle =$ L.C. of $|1\rangle$ & $|2\rangle$

$$\tilde{A} = \lambda \left(\begin{array}{cc|c} 0 & 1 & 0 \\ 1 & 0 & 0 \\ \hline 0 & 0 & 2 \end{array}\right)$$

$\tilde{A}$ is diagonal in the subspace spanned by $|2\rangle$.

Hence, eigenvalue of $\tilde{A}$ is 2λ with eigenket

$$|2\lambda\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}.$$

And in the subspace spanned by $|0\rangle$ and $|1\rangle$, we have

$$\tilde{A} = \lambda \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

So eigenvalues are $(0+1)\lambda$ and $(0-1)\lambda$ with eigenkets

$$|\pm\lambda\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ \pm 1 \end{pmatrix}$$

So the eigenkets in the complete computational basis is

$$|\pm\lambda\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ \pm 1 \\ 0 \end{pmatrix}$$

Similarly,

$$\tilde{B} = \mu \left(\begin{array}{c|cc} 2 & 0 & 0 \\ \hline 0 & 0 & 1 \\ 0 & 1 & 0 \end{array}\right)$$

$\tilde{B}$ is diagonal in the subspace spanned by $|0\rangle$.

Hence, eigenvalue of $\tilde{B}$ is 2μ with eigenket

$$|2\mu\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}.$$

In the subspace spanned by $|1\rangle$ and $|2\rangle$, we have

$$\tilde{B} = \mu \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

So eigenvalues are $(0+1)\mu$ and $(0-1)\mu$ with eigenkets

$$|\pm\mu\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ \pm 1 \end{pmatrix}$$

In the complete computational basis, the eigenkets are

$$|\pm\mu\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ \pm 1 \end{pmatrix}$$

(b) Suppose the system starts out in the generic state

$$|S(0)\rangle = \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix},$$

with $|c_1|^2 + |c_2|^2 + |c_3|^2 = 1$. Find the expectation values (at $t=0$) of H, A, and B.

To calculate the expectation value at time,

1. we need to find wavefunction at time t i.e. $|\Psi(t)\rangle$

2. $\langle Q \rangle = \langle \Psi(t) | \tilde{Q} | \Psi(t) \rangle$

Now

1. $|\Psi(t)\rangle = e^{-i\frac{\tilde{H}t}{\hbar}} |\Psi(0)\rangle$

The exponential operator $e^{-i\frac{\tilde{H}t}{\hbar}}$ can be shown to ~~be~~ have the diagonal matrix representation

$$e^{-\frac{i\tilde{H}t}{\hbar}} = \begin{pmatrix} e^{-\frac{iE_0 t}{\hbar}} & 0 & 0 \\ 0 & e^{-\frac{iE_1 t}{\hbar}} & 0 \\ 0 & 0 & e^{-\frac{iE_2 t}{\hbar}} \end{pmatrix}$$

$$E_0 = \hbar\omega, \quad E_1 = E_2 = 2\hbar\omega$$

So $$e^{-i\frac{\tilde{H}t}{\hbar}} = \begin{pmatrix} e^{-i\omega t} & 0 & 0 \\ 0 & e^{-i2\omega t} & 0 \\ 0 & 0 & e^{-i2\omega t} \end{pmatrix}$$

so $$|\Psi(t)\rangle = \begin{pmatrix} e^{-i\omega t} & 0 & 0 \\ 0 & e^{-i2\omega t} & 0 \\ 0 & 0 & e^{-i2\omega t} \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix}$$

$$= \begin{pmatrix} c_1 e^{-i\omega t} \\ c_2 e^{-i2\omega t} \\ c_3 e^{-i2\omega t} \end{pmatrix}$$

so that

$$\langle \Psi(t)| = \left(c_1^* e^{+i\omega t} \quad c_2^* e^{+i2\omega t} \quad c_3^* e^{+i2\omega t} \right)$$

$$\therefore 2.\ \langle \tilde{H} \rangle = \langle \psi(t) | H | \psi(t) \rangle$$

$$= \langle \psi(t) | \begin{pmatrix} \hbar\omega & 0 & 0 \\ 0 & 2\hbar\omega & 0 \\ 0 & 0 & 2\hbar\omega \end{pmatrix} \begin{pmatrix} c_1 e^{-i\omega t} \\ c_2 e^{-i2\omega t} \\ c_3 e^{-i2\omega t} \end{pmatrix}$$

$$= \langle \psi(t) | \begin{pmatrix} c_1 \hbar\omega e^{-i\omega t} \\ c_2 2\hbar\omega e^{-i2\omega t} \\ c_3 2\hbar\omega e^{-i2\omega t} \end{pmatrix}$$

$$= \begin{pmatrix} c_1^* e^{+i\omega t} & c_2^* e^{+i2\omega t} & c_3^* e^{+i2\omega t} \end{pmatrix} \begin{pmatrix} c_1 \hbar\omega e^{-i\omega t} \\ c_2 2\hbar\omega e^{-i2\omega t} \\ c_3 2\hbar\omega e^{-i2\omega t} \end{pmatrix}$$

$$= |c_1|^2 \hbar\omega + |c_2|^2 2\hbar\omega + |c_3|^2 2\hbar\omega$$

$$\boxed{\therefore \langle \tilde{H} \rangle = \hbar\omega \left(|c_1|^2 + 2|c_2|^2 + 2|c_3|^2 \right)}$$

Now

$$\langle \tilde{A} \rangle = \langle \psi(t) | \tilde{A} | \psi(t) \rangle$$

$$= \langle \psi(t) | \begin{pmatrix} 0 & \lambda & 0 \\ \lambda & 0 & 0 \\ 0 & 0 & 2\lambda \end{pmatrix} \begin{pmatrix} c_1 e^{-i\omega t} \\ c_2 e^{-i2\omega t} \\ c_3 e^{-i2\omega t} \end{pmatrix}$$

$$= \langle \psi(t) | \begin{pmatrix} \lambda c_2 e^{-i2\omega t} \\ \lambda c_1 e^{-i\omega t} \\ 2\lambda c_3 e^{-i2\omega t} \end{pmatrix}$$

$$= \begin{pmatrix} c_1^* e^{+i\omega t} & c_2^* e^{+i2\omega t} & c_3^* e^{+i2\omega t} \end{pmatrix} \begin{pmatrix} \lambda c_2 e^{-i2\omega t} \\ \lambda c_1 e^{-i\omega t} \\ 2\lambda c_3 e^{-i2\omega t} \end{pmatrix}$$

$$= \lambda c_1^* c_2 e^{-i\omega t} + \lambda c_2^* c_1 e^{+i\omega t} + 2\lambda |c_3|^2$$

$$= \lambda c_1^* c_2 e^{-i\omega t} + \lambda (c_1^* c_2 e^{-i\omega t})^* + 2\lambda |c_3|^2$$

$$\boxed{\therefore \langle \tilde{A} \rangle = 2\lambda \, \mathrm{Re}(c_1^* c_2 e^{-i\omega t}) + 2\lambda |c_3|^2}$$

Similarly,

$$\langle B \rangle = \langle \psi(t) | \tilde{B} | \psi(t) \rangle$$

$$= \langle \psi(t) | \begin{pmatrix} 2\mu & 0 & 0 \\ 0 & 0 & \mu \\ 0 & \mu & 0 \end{pmatrix} \begin{pmatrix} c_1 e^{-i\omega t} \\ c_2 e^{-i2\omega t} \\ c_3 e^{-i2\omega t} \end{pmatrix}$$

$$= \langle \psi(t) | \begin{pmatrix} 2 c_1 \mu e^{-i\omega t} \\ c_3 \mu e^{-i2\omega t} \\ c_2 \mu e^{-i2\omega t} \end{pmatrix}$$

$$= \begin{pmatrix} c_1^* e^{+i\omega t} & c_2^* e^{+i2\omega t} & c_3^* e^{+i2\omega t} \end{pmatrix} \begin{pmatrix} 2\mu c_1 e^{-i\omega t} \\ \mu c_3 e^{-i2\omega t} \\ \mu c_2 e^{-i2\omega t} \end{pmatrix}$$

$$= 2\mu |c_1|^2 + \mu \, c_2^* c_3 e^0 + c_3^* c_2 e^0$$

$$\boxed{\therefore \langle B \rangle = 2\mu |c_1|^2 + 2\mu \, \mathrm{Re}(c_2^* c_3)}$$

Observations:

1. $\langle H \rangle$ is independent of time.
2. $\langle A \rangle$ is function of time.
3. $\langle B \rangle$ is independent of time.

(c) What is $|S(t)\rangle$? If you measured the energy of this state (at time t), what values might you get, and what ~~is~~ are the probabilities of each? Answer the same questions for observables A and B.

The state at time t is

$$|S(t)\rangle = \begin{pmatrix} c_1 e^{-i\omega t} \\ c_2 e^{-2i\omega t} \\ c_3 e^{-2i\omega t} \end{pmatrix}$$

The Hamiltonian has eigenvalues :

$E_0 = \hbar\omega$ with degeneracy $= 1$

$E_1, E_2 = 2\hbar\omega$ with degeneracy $= 2$

So two measurement outcomes are possible –

$E = \hbar\omega, 2\hbar\omega$

and probability to get $E = \hbar\omega$ is

$$|\langle \hbar\omega | \psi(t)\rangle|^2 = |\langle 0 | \psi(t)\rangle|^2$$
$$= |c_1 e^{-i\omega t}|^2$$
$$= |c_1|^2$$

and probability to get $E = 2\hbar\omega$ is

$$|\langle 1 | \psi(t)\rangle|^2 + |\langle 2 | \psi(t)\rangle|^2$$
$$= |c_2|^2 + |c_3|^2$$

Similarly,
for $\tilde{A}$, the ~~eigenvalues~~ outcomes are

$$-\lambda\ ,\ +\lambda\ ;\ +2\lambda$$

and probabilities are

$$|\langle -\lambda|\psi(t)\rangle|^2,\ |\langle +\lambda|\psi(t)\rangle|^2,\ |\langle +2\lambda|\psi(t)\rangle|^2$$

i.e. $$\left|\frac{1}{\sqrt{2}}\begin{pmatrix}1\\-1\\0\end{pmatrix}^{\dagger}\begin{pmatrix}c_1e^{-i\omega t}\\c_2e^{-i2\omega t}\\c_3e^{-i2\omega t}\end{pmatrix}\right|^2,\ \left|\frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\\0\end{pmatrix}^{\dagger}\begin{pmatrix}c_1e^{-i\omega t}\\c_2e^{-i2\omega t}\\c_3e^{-i2\omega t}\end{pmatrix}\right|^2,\ \left|\begin{pmatrix}0\\0\\1\end{pmatrix}^{\dagger}\begin{pmatrix}\vdots\end{pmatrix}\right|^2$$

i.e. $$\left|\frac{1}{\sqrt{2}}\left(c_1e^{-i\omega t}-c_2e^{-i2\omega t}\right)\right|^2,\ \left|\frac{1}{\sqrt{2}}\left(c_1e^{-i\omega t}+c_2e^{-i2\omega t}\right)\right|^2,\ |c_3|^2$$

i.e. $$\frac{1}{2}(|c_1|^2+|c_2|^2)-c_1c_2\cos\omega t,$$

$$\frac{1}{2}(|c_1|^2+|c_2|^2)+c_1c_2\cos\omega t,$$

$|c_3|^2$ respectively

Similarly
for $\tilde{B}$, the ~~eigenvalues~~ outcomes are

$$-\mu\ ,\ +\mu\ ,\ +2\mu$$

with probabilities

$$|\langle -\mu|\psi(t)\rangle|^2,\ |\langle +\mu|\psi(t)\rangle|^2,\ |\langle +2\mu|\psi(t)\rangle|^2$$

$$\frac{1}{2}(|c_2|^2+|c_3|^2),\ \frac{1}{2}(|c_2|^2+|c_3|^2),\ |c_1|^2$$

	Measurement outcome ~~at time t~~	probability
$\tilde{H}$	$\hbar\omega$	$\lvert c_1\rvert^2$
	$2\hbar\omega$	$\lvert c_2\rvert^2+\lvert c_3\rvert^2$
$\tilde{A}$	$-\lambda$	$\frac{1}{2}(\lvert c_1\rvert^2+\lvert c_2\rvert^2)-c_1c_2\cos\omega t$
	$+\lambda$	$\frac{1}{2}(\lvert c_1\rvert^2+\lvert c_2\rvert^2)+c_1c_2\cos\omega t$
	$+2\lambda$	$\lvert c_3\rvert^2$
$\tilde{B}$	$-\mu$	$\frac{1}{2}(\lvert c_2\rvert^2+\lvert c_3\rvert^2)$
	$+\mu$	$\frac{1}{2}(\lvert c_2\rvert^2+\lvert c_3\rvert^2)$
	$+2\mu$	$\lvert c_1\rvert^2$

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END OF ASSIGNMENT